
Piano Piece Difficulty Estimator

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Introduction

Piano students often spend considerable time searching for repertoire that is challenging but still playable [1]. Difficulty ratings from exam syllabi such as RCM [2] help this process and give students and educators a way to plan progression. However, these ratings cover mainly a limited subset of classical music, leaving new or lesser-known repertoire without reliable difficulty labels. This project aims to develop a deep learning model to predict a continuous difficulty level of a piano piece from a MIDI file, a popular format to represent symbolic music. The model could help students find music, teachers recommend repertoire, and composers or arrangers estimate the playability of their piece. Deep learning is a suitable approach for this task due to the complex nature of piano pieces and the fact that difficulty depends on many features and the non-obvious relationships between them [3].

Illustration

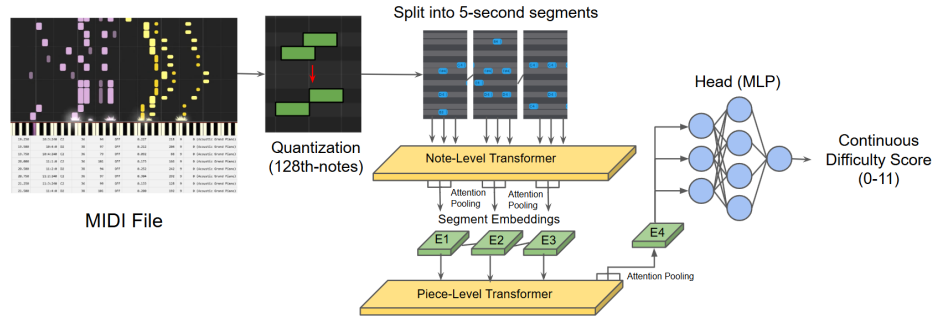


Figure 1: Primary model hierarchical transformer architecture

Background & Related Work

Piano difficulty estimation has been studied before using various feature-based and deep learning approaches. Chiu and Chen [4] used a selection algorithm to extract musical features directly from MIDI data, then employed regression-based approaches to estimate difficulties. Later methods have shifted toward deep learning to better capture complex musical relationships.

Ghatas et al. [3] used a matrix-based visual approach by converting from MIDI to a 2-dimensional piano roll representation of music, facilitating use of CNNs and improving accuracy over previous works when combined with handcrafted feature analysis.

Ramonedá et al. [5] targeted the problem through computed features associated with fingerings. These were then used as input into a GRU with an attention mechanism and gradient-boosted trees to estimate difficulty on both local score segments and the whole piece. To bypass the scarcity of machine-readable scores,

Ramonedá et al. [6] estimated difficulty from sheet music images directly by extracting the most relevant features of the sheet music into a "bootleg score", focusing only on the positions of the noteheads and staff lines, then using a GPT-based model.

Zhang et al. [7] systematically evaluated various symbolic music formats used for music classification tasks, namely matrix (piano roll), sequence (MIDI), and graph representations. Using common architectures for each representation, they found that each format yields comparable classification accuracy, with graphs emerging as a promising alternative.

Data Processing

Step	Processing Done	Purpose / Output
Get raw dataset	Downloaded PianoSyllabus metadata and MIDI files for 7901 pieces [8].	Main supervised dataset with difficulty labels from 0–10.
Metadata cleaning	Removed accents from titles and kept only useful fields: title, composer, difficulty, MIDI path, and chapter data.	Cleaner metadata for matching and model preparation.
MIDI validation	Checked that each MIDI path in the metadata exists and could be parsed.	Removed unusable or corrupted training inputs.
Artifact removal	Removed MIDI files that were less than 30 seconds because of erroneous extraction of an intro chapter.	Removed 6 short intro fragments that were not real full-piece examples.
Movement/title validation	Manually removed pieces where MIDI content did not match the title due to incorrect chapter selection.	Removed 62 pieces, resulting in 7833 clean examples.
PianoLibrary scraping	Described in main body text below.	Changed 327 pieces to difficulty 11.
Dataset split	Split by composer into roughly 70% train, 15% validation, 15% test.	Produced 5488 train, 1175 validation, and 1170 test examples.

Table 1: Summary of completed data cleaning steps

Steps 4 and 5 in Table 1 refer to a known issue with the dataset: many PianoSyllabus MIDI files were originally created by extracting a portion of a YouTube video using chapter timestamps [8]. This sometimes caused the dataset to include a short artifacted intro chapter rather than the full piece, or only one movement of a multi-movement work, which had to be removed.

Another major issue was that many advanced pieces were assigned the maximum label of 10, which encompassed a wide difficulty range. To reduce this ceiling effect, difficulty ratings from 0-5 from PianoLibrary [9] were scraped and matched to PianoSyllabus difficulty-10 pieces using composer name, fuzzy title similarity, catalogue number, key, movement number, and form (e.g. sonata, etude) to build a similarity score. After manual validation, matched pieces with PianoLibrary difficulty of 4 or higher were adjusted to a new difficulty label of 11 while the remainder stayed at difficulty 10.

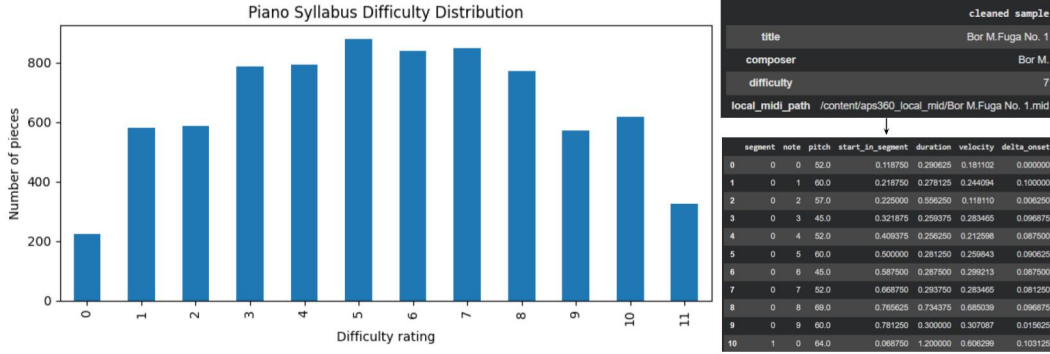


Figure 2: Statistics and cleaned example for the final dataset

Baseline Model

The baseline model is a random forest regressor that predicts based on on handcrafted MIDI features of the piece, namely piece duration, number of notes, notes per second, pitch range, mean and standard deviation of note duration, mean and maximum pitch jump, and mean tempo. This is a reasonable baseline because it uses a simple method, trains quickly, and requires limited tuning. With 200 estimators and minimum leaf size of 3 the baseline model achieved a test exact accuracy of 28.7% and a mean absolute error (MAE) of 1.161 on the test dataset. Inspection showed that the baseline often overestimates easy but long pieces and underestimates hard but short pieces. This suggests that the model mostly relies on broad dataset correlations, limiting the maximum accuracy it can achieve.

Primary Model

The primary model uses a hierarchical Transformer architecture. MIDI notes are quantized, given positional encodings, and processed within short 5-second segments, with segment embeddings being combined to represent the full piece (Figure 1). Each note is represented by pitch, onset within the segment, duration, velocity, and time from last onset. Since each batch needs fixed-size tensors while MIDI files have variable lengths, segments are padded to have the same number of notes, and pieces within each batch are padded to have the same number of segments. Pieces are sorted by length and taken in order when forming batches to reduce the required padding, speeding up training.

The final model uses 2 note-level and 2 piece-level Transformer layers, each with 64-dimensional embeddings, 4 attention heads, feedforward dimension 192, and dropout 0.1, resulting in 175,523 trainable parameters. Attention pooling is used after both Transformer levels, and the final piece embedding is passed through an MLP with one hidden layer of size 64 to predict a continuous difficulty score. ReLU was used as the activation function throughout. The best model uses AdamW with a batch size of 16, learning rate of $5e-4$, weight decay of $1e-4$, and was trained for 30 epochs. To address class imbalance, the MSE loss was weighted by each class’s inverse square-root frequency.

Quantitative Results

Metric	Value	Description / Importance
Validation accuracy	33.79%	Percentage of predictions in the validation set that exactly matched the true difficulty from 0–11 after rounding.
Test accuracy	31.45%	Percentage of predictions in the test set that exactly matched the true difficulty from 0–11 after rounding.
Macro average test accuracy	31.66%	Shows that class imbalance is not affecting the testing much.
Within-1 accuracy (test)	77.61%	Percentage of rounded predictions within one level of the true difficulty. Much higher than exact accuracy, showing that the model effectively differentiates pieces but is imprecise.
Within-2 accuracy (test)	94.10%	Percentage of rounded predictions within two levels of the true difficulty. Indicates that the model rarely has large errors.
Mean absolute error (test)	1.009	Average absolute difference between the raw continuous prediction and the true difficulty.

Table 2: Quantitative results for the best Transformer model.

Qualitative Results

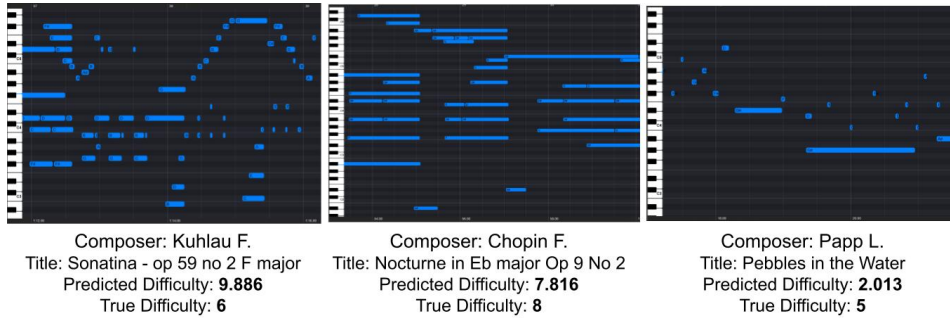


Figure 3: Piano roll representations of three example inputs and outputs

Three pieces were selected to show strengths and limitations of the model. *Sonatina* was overestimated: the model appropriately recognized its fast runs and high note density as signs of technical complexity, but in reality, many passages are scales or arpeggios with convenient fingerings. This suggests that the model can identify markers of difficulty but lacks an understanding of how notes actually fit under the hand. *Nocturne* was correctly predicted as 8 (relatively high) despite its low tempo and note density, hinting that the model can recognize the specific difficult patterns such as left-hand jumps and melodic runs. *Pebbles in the Water* was underestimated because its difficulty comes from precise staccato timing, light touch, and control of dynamics, indicating weaker performance when difficulty depends on expressive technique or interpretation rather than the notes themselves.

Evaluation On New Data

Since PianoSyllabus already contains most pieces labelled using exam syllabi such as RCM, it is infeasible to find more data using the same 0–11 scale. Instead, 180 MIDI files (45 of each difficulty) were scraped from 8notes.com [10], labeled with 4 difficulty categories and containing genres such as jazz, rock, and pop, making it a good test of the model’s ability to generalize. Pieces were validated to ensure that only previously unseen composers were included. Since these categories do not directly correspond to the model’s difficulty scale, ranking metrics were used instead of accuracy.

Dataset	Pairwise Ranking Accuracy	Spearman Correlation	Kendall τ_b
8notes.com dataset	89.15%	0.824	0.680
PianoSyllabus test set*	93.58%	0.880	0.750

Table 3: Ordinal performance metrics on 8notes and the bucketed PianoSyllabus test set.

To make the ranking metrics more comparable, the PianoSyllabus test labels were grouped into 0–2, 3–5, 6–8, and 9–11, although these ranges do not exactly match the independently defined 8notes categories. The 89.15% pairwise ranking accuracy is well above the 50% expected from random ordering, showing that the model usually ranked the more difficult piece higher. The Spearman correlation of 0.824 and Kendall’s τ_b of 0.680 are also relatively high, indicating strong agreement between the predictions and the overall difficulty ordering. The decrease between the two datasets was expected and is likely due to the different difficulty scale and methodology, broadening of genres, and possible differences between performance-derived and score-like MIDI timing.

Discussion

The model is overall effective for practical difficulty estimation, especially given the challenge of predicting a difficulty across 12 levels even when neighboring levels may differ only slightly. The labels are also imperfect because difficulty is subjective: exam systems and reviewers may weight factors like technical difficulty and interpretation differently. The same piece may also feel easier or harder depending on a pianist’s hand size and experience. The high within-one and within-two accuracies therefore provide a more realistic measure of usefulness and show that the predictions are close enough to support repertoire selection and estimating the playability of new compositions.

The baseline performed better than expected, showing that broad features such as tempo, note density, duration, and pitch jumps already capture much of what makes a piece difficult. The Transformer still improved on this, suggesting that it was able to learn useful information from specific note patterns and how difficulty develops across passages. Its performance on unseen data also demonstrated that these patterns generalize well. Some remaining errors may come from residual dataset issues, such as incorrect MIDI transcription or inconsistent labels, despite the cleaning process. The model also showed some overfitting with validation accuracy peaking at epoch 8 while training accuracy continued increasing, so more data would likely improve its performance, especially for sparse classes such as 0 and 11. Some errors are also difficult to avoid because MIDI does not directly represent fingering, hand assignment, or explicit dynamics and articulations, all of which affect difficulty.

An interesting finding was that changes in the input representation, such as quantization and provided note features, often had a greater effect than hyperparameter changes such as batch size or embedding dimension. Notably, adding a "delta onset" feature (the time since the previous note onset) noticeably improved performance, even though it could already be derived from the note start times, showing that directly providing useful shortcuts can make it easier for the model to learn. Other experiments explored different optimizers, loss functions, and activation functions, changing segment length, using max pooling, and a chord-size feature, but these only produced small or inconsistent gains.

Ethical Considerations

This project relies on MIDI files of piano recordings, which may be copyrighted. Thus, data should only be used for research purposes and not redistributed. Scraped data may also be subject to website terms, and any future collection should use rate limits and caching to avoid placing unnecessary load on source websites. As piano difficulty is subjective and may not generalize to all pianists, the model’s predictions may mislead learners into attempting repertoire beyond their ability and risk injury or discouragement. Also, as the model is mainly trained on classical music, it may be less accurate for other styles, possibly causing a bias towards certain genres.

References

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